

**IEEE VTC-Spring 2026: The 3rd Workshop on
Synergizing Digital Twins and Pervasive Intelligence for Next-Generation IoT/IoV
Call for Papers**

Next-generation Internet of Things (IoT) and Internet of Vehicles (IoV) systems are evolving toward highly autonomous, hyper-connected, and intelligence-driven ecosystems. Future networking, computing, and control infrastructures must support massive-scale connectivity, high mobility, heterogeneous services, and ultra-reliable low-latency communications across diverse IoT/IoV scenarios, including connected and autonomous vehicles, intelligent transportation systems, roadside and aerial sensing, and edge–cloud collaborative environments. These systems face fundamental challenges arising from dynamic and large-scale network topologies, rapidly time-varying wireless conditions, intermittent connectivity, spectrum scarcity, ultra-dense deployments, and stringent safety, reliability, and latency requirements for mission-critical applications. Digital twins (DTs) have emerged as a powerful paradigm for managing the complexity of IoT/IoV systems by providing virtual representations of physical entities, networks, and environments that continuously reflect system states and dynamics. When combined with pervasive artificial intelligence (AI), embedded across devices, vehicles, edge nodes, and cloud platforms, DTs enable predictive analysis, autonomous decision-making, and closed-loop optimization across physical and virtual domains. This DT-AI synergy supports proactive and self-optimizing IoT/IoV operation, including intelligent resource management, context-aware routing and scheduling, adaptive network slicing, cooperative perception and control, predictive maintenance, and resilient system adaptation envisioned for 6G-enabled IoT/IoV ecosystems. Despite their promise, integrating DTs and AI in large-scale and highly dynamic IoT/IoV systems raises significant challenges, such as scalable DT modeling and synchronization under mobility and partial observability, efficient data collection and fusion across heterogeneous domains, latency- and energy-constrained AI inference at the edge, and ensuring trustworthy, secure, and privacy-preserving DT–AI operations. Addressing these challenges requires new architectures, protocols, and cross-layer optimization frameworks that jointly consider sensing, communication, computing, and intelligence.

This workshop invites high-quality original research contributions that explore theories, architectures, algorithms, and experimental systems for synergizing digital twins and pervasive intelligence in next-generation IoT/IoV. Both theoretical advances and practical insights from testbeds, prototypes, and field trials are encouraged.

- DT architectures and modeling techniques for IoT and IoV systems
- AI-enhanced digital twins for prediction, optimization, and autonomous control
- Joint DT-AI frameworks for real-time IoT/IoV network and system optimization
- Pervasive intelligence in self-organizing, adaptive, and autonomous IoT/IoV systems
- Edge-cloud collaboration for DT-enabled intelligence in IoT/IoV
- Distributed, federated, and privacy-preserving learning for DT-AI integration
- Resource management, routing, scheduling, and network slicing enabled by DT-AI
- Security, privacy, and trust in DT-AI-enabled IoT/IoV ecosystems
- Interoperability and standardization of DT and AI technologies for IoT/IoV and 6G
- Big data management and real-time analytics for large-scale digital twins
- DT-AI-enabled applications in intelligent transportation and smart infrastructure
- UAV/drone-assisted IoT/IoV sensing and monitoring using DT and AI
- Experimental platforms, testbeds, and field trials for DT-AI-enabled IoT/IoV
- Cross-domain innovations enabled by DT-AI synergy in smart mobility and beyond

Paper submission: Papers must be formatted in the standard IEEE two-column format that is used by the VTC-Spring 2026 main conference and must not exceed six pages in length (including references).

Workshop Organization Committee:

General Co-chairs

Trung Q. Duong, Memorial University, Canada
Tom H. Luan, Xi'an Jiaotong University, China
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Publicity Co-chairs

Jie Gao, Carleton University, Canada
Nan Chen, University of Ottawa, Canada
Omar Alhussein, Khalifa University, UAE

Important Dates:

Paper Submission Deadline: 25 January 2026
Paper Acceptance Notification: 18 March 2026
Final Paper Submission: 1 April 2026